Enums

The last class-like construct in PHP we're going to look at is the *enum*. The enum is PHP's implementation of an enumerated type, which in general terms is a data type consisting of a fixed set of values. These can be helpful when you want to restrict the values of things to valid states.

One of the classic examples of an enum type uses the suits of a deck of cards. We can represent the suit of a card with an enum consisting of four strings: "Clubs", "Diamonds", 'Hearts', and "Spades". If a `Card` class exists in your program, you can restrict its `::$suit` property to one of the four enum values (and never have to worry about accidentally breaking code if someone tries to play the four of cabbages).

Prior to the introduction of the `enum` type in PHP 8.1, this was usually achieved with class constants and would probably look something like this:

```
class Suit
{
    public const CLUBS = 'clubs';
    public const DIAMONDS = 'diamonds';
    public const HEARTS = 'hearts';
    public const SPADES = 'spades';
}

class Card
{
    public function __construct(
        private readonly string $suit,
        private readonly string $value
    ) {
    }

    public function getFullName(): string
    {
        return "{$this->value} of {$this->suit}";
    }
}

$card = new Card(Suit::SPADES, "ace");
print "The {$card->getFullName()}, the {$card->getFullName()} \m/" . PHP_EOL;
```